gas cloud to the earth's demise when the sun swells into a red giant. It's all there.

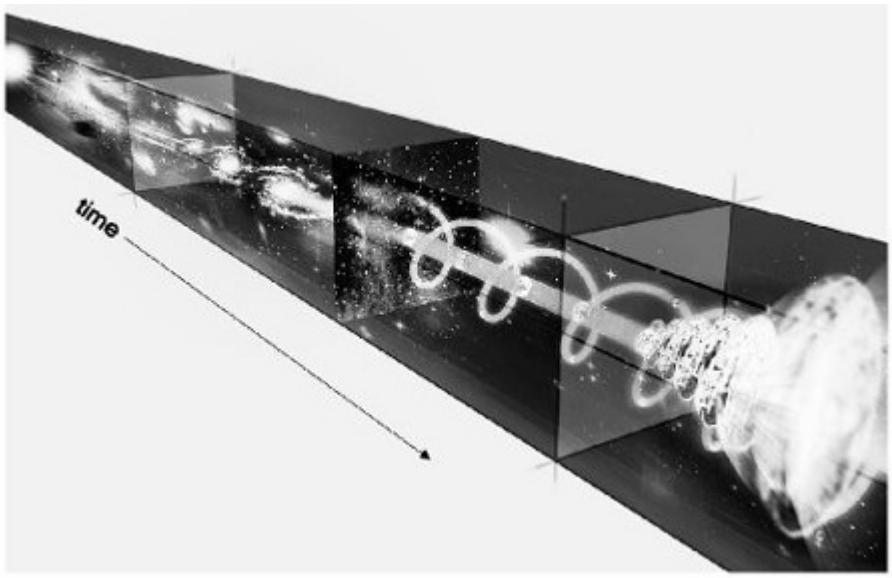


Figure 5.1 A schematic depiction of all space throughout all time (depicting, of course, only part of space through part of time) showing the formation of some early galaxies, the formation of the sun and the earth, and the earth's ultimate demise when the sun swells into a red giant, in what we now consider our distant future.

Unquestionably, Figure 5.1 is an imaginary perspective. It stands outside of space and time. It is the view from nowhere and nowhen. Even so—even though we can't actually step beyond the confines of spacetime and take in the full sweep of the universe—the schematic depiction of Figure 5.1 provides a powerful means of analyzing and clarifying basic properties of space and time. As a prime example, the intuitive sense of time's flow can be vividly portrayed in this framework by a variation on the movie-projector metaphor. We can envision a light that illuminates one time slice after another, momentarily making the slice come alive in the present—making it the momentary *now*—only to let it go instantly dark again as the light moves on to the next slice. Right now, in this intuitive way of thinking about time, the light is illuminating the slice in which you, sitting on planet earth, are reading *this* word, and now it is illuminating the slice in which you are reading *this* word. But, again, while this image seems to match experience, scientists have been unable to find anything in the laws of physics that embodies such a moving light. They have found no physical mechanism that singles out